

Complete Mathematical Proof of GHZ-Based Quantum Visual Secret Sharing

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1 Classical Image Representation

We define two binary images:

$$G_1 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad G_2 = \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}$$

Each pixel is indexed by (i, j) :

$$(g_{1,ij}, g_{2,ij}) \in \{0, 1\}^2$$

Pixels with the same index from both images are paired as follows:

$$(g_{1,11}, g_{2,11}) = (0, 0)$$

$$(g_{1,12}, g_{2,12}) = (1, 0)$$

$$(g_{1,21}, g_{2,21}) = (1, 1)$$

$$(g_{1,22}, g_{2,22}) = (0, 1)$$

Each pair will be encoded into a 3-qubit quantum state.

2 Single-Qubit Foundations

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

2.1 X Gate

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

2.2 Hadamard Gate

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

3 GHZ State Construction

Initial state:

$$|000\rangle$$

3.1 Apply Hadamard to Qubit 0

$$(H \otimes I \otimes I)|000\rangle$$

Compute:

$$H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

So:

$$= \frac{1}{\sqrt{2}}(|000\rangle + |100\rangle)$$

3.2 Apply First CNOT (0 $\rightarrow$ 1)

$$|000\rangle \rightarrow |000\rangle$$

$$|100\rangle \rightarrow |110\rangle$$

Thus:

$$\frac{1}{\sqrt{2}}(|000\rangle + |110\rangle)$$

3.3 Apply Second CNOT (0 $\rightarrow$ 2)

$$|000\rangle \rightarrow |000\rangle$$

$$|110\rangle \rightarrow |111\rangle$$

Final GHZ state:

$$|\text{GHZ}\rangle = \frac{1}{\sqrt{2}}(|000\rangle + |111\rangle)$$

4 Encoding Pixel Values

We apply:

$$I \otimes X^{g_1} \otimes X^{g_2}$$

Using identity:

$$X^0 = I, \quad X^1 = X$$

Also:

$$X|b\rangle = |b \oplus 1\rangle$$

Thus:

$$X^g|b\rangle = |b \oplus g\rangle$$

4.0.1 To prove $X^g|b\rangle = |b \oplus g\rangle$

We go through 4 possible cases:

g	b	X^g	$X^g b\rangle$	$ b \oplus g\rangle$
0	0	I	$I 0\rangle = 0\rangle$	$ 0 \oplus 0\rangle = 0\rangle$
0	1	I	$I 1\rangle = 1\rangle$	$ 1 \oplus 0\rangle = 1\rangle$
1	0	X	$X 0\rangle = 1\rangle$	$ 0 \oplus 1\rangle = 1\rangle$
1	1	X	$X 1\rangle = 0\rangle$	$ 1 \oplus 1\rangle = 0\rangle$

Table 1: Proving of the identity $X^g|b\rangle = |b \oplus g\rangle$ for all possible values of $g, b \in \{0, 1\}$.

4.1 Apply to GHZ

$$|\psi_{g_1, g_2}\rangle = (I \otimes X^{g_1} \otimes X^{g_2}) \frac{1}{\sqrt{2}}(|000\rangle + |111\rangle)$$

4.2 Distribution

$$\begin{aligned} & (I \otimes X^{g_1} \otimes X^{g_2}) \frac{1}{\sqrt{2}}(|000\rangle + |111\rangle) \\ &= \frac{1}{\sqrt{2}} \left[(I \otimes X^{g_1} \otimes X^{g_2})|000\rangle + (I \otimes X^{g_1} \otimes X^{g_2})|111\rangle \right] \end{aligned}$$

Using:

$$(A \otimes B \otimes C)(x \otimes y \otimes z) = (Ax) \otimes (By) \otimes (Cz)$$

g_1	g_2	First Term: $ 0, g_1, g_2\rangle$	Second Term: $ 1, 1 \oplus g_1, 1 \oplus g_2\rangle$	Final State
0	0	$ 000\rangle$	$ 111\rangle$	$\frac{1}{\sqrt{2}}(000\rangle + 111\rangle)$
0	1	$ 001\rangle$	$ 110\rangle$	$\frac{1}{\sqrt{2}}(001\rangle + 110\rangle)$
1	0	$ 010\rangle$	$ 101\rangle$	$\frac{1}{\sqrt{2}}(010\rangle + 101\rangle)$
1	1	$ 011\rangle$	$ 100\rangle$	$\frac{1}{\sqrt{2}}(011\rangle + 100\rangle)$

Table 2: Proving of GHZ state distribution under the operation $I \otimes X^{g_1} \otimes X^{g_2}$ for all possible values of $g_1, g_2 \in \{0, 1\}$.

Thus:

$$|\psi_{g_1, g_2}\rangle = \frac{1}{\sqrt{2}}(|0, g_1, g_2\rangle + |1, 1 \oplus g_1, 1 \oplus g_2\rangle)$$

5 Measurement Outcomes

Measurement yields:

$$(r_0, r_1, r_2) = \begin{cases} (0, g_1, g_2) \\ (1, 1 \oplus g_1, 1 \oplus g_2) \end{cases}$$

5.1 Why Only Two Outcomes

The state contains only two basis states $\rightarrow$ only those have non-zero probability.

6 Pixel-by-Pixel Computation

6.1 Pixel (1,1): (0,0)

State:

$$\frac{1}{\sqrt{2}}(|000\rangle + |111\rangle)$$

Measured State	r_0	r_1	r_2
000	0	0	0
111	1	1	1

Table 3: Possible measurement outcomes for Pixel (1,1) with secret bits $(g_1, g_2) = (0, 0)$, corresponding to the GHZ state $\frac{1}{\sqrt{2}}(|000\rangle + |111\rangle)$.

6.2 Pixel (1,2): (1,0)

$$\begin{aligned} X \otimes I : |000\rangle &\rightarrow |010\rangle \\ |111\rangle &\rightarrow |101\rangle \end{aligned}$$

State:

$$\frac{1}{\sqrt{2}}(|010\rangle + |101\rangle)$$

Measured State	r_0	r_1	r_2
010	0	1	0
101	1	0	1

Table 4: Possible measurement outcomes for Pixel (1,2) with secret bits $(g_1, g_2) = (1, 0)$, corresponding to the GHZ state $\frac{1}{\sqrt{2}}(|010\rangle + |101\rangle)$.

6.3 Pixel (2,1): (1, 1)

State:

$$\frac{1}{\sqrt{2}}(|011\rangle + |100\rangle)$$

Measured State	r_0	r_1	r_2
011	0	1	1
100	1	0	0

Table 5: Possible measurement outcomes for Pixel (2,1) with secret bits $(g_1, g_2) = (1, 1)$, corresponding to the GHZ state $\frac{1}{\sqrt{2}}(|011\rangle + |100\rangle)$.

6.4 Pixel (2,2): (0, 1)

State:

$$\frac{1}{\sqrt{2}}(|001\rangle + |110\rangle)$$

Measured State	r_0	r_1	r_2
001	0	0	1
110	1	1	0

Table 6: Possible measurement outcomes for Pixel (2,2) with secret bits $(g_1, g_2) = (0, 1)$, corresponding to the GHZ state $\frac{1}{\sqrt{2}}(|001\rangle + |110\rangle)$.

7 Constructing Share Matrices

As there are two possible outcomes of each measurement, there are a total of 16 outcomes. We will be discussing 2 outcomes in the following subsections, specifically the First Outcome (all first rows of the tables) and the Second Outcome (all second rows of the tables).

7.1 First Outcome

$$U = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \quad S_1 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad S_2 = \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}$$

7.2 Second Outcome

$$U = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \quad S_1 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad S_2 = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$$

8 Reconstruction of the Original Images

The reconstruction process uses the relations:

$$g_{1,ij} = r_{0,ij} \oplus r_{1,ij}, \quad g_{2,ij} = r_{0,ij} \oplus r_{2,ij}$$

We now reconstruct each pixel individually using the previously obtained measurement results.

8.1 Given First Outcome

$$U = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}, \quad S_1 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad S_2 = \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}$$

These correspond to:

$$U = [r_{0,ij}], \quad S_1 = [r_{1,ij}], \quad S_2 = [r_{2,ij}]$$

8.1.1 Pixel-by-Pixel Reconstruction

Pixel Pair (i, j)	$g_{1,ij} = r_{0,ij} \oplus r_{1,ij}$	$g_{2,ij} = r_{0,ij} \oplus r_{2,ij}$
$(1, 1)$	$g_{1,11} = r_{0,11} \oplus r_{1,11}$ $= 0 \oplus 0$ $= 0$	$g_{2,11} = r_{0,11} \oplus r_{2,11}$ $= 0 \oplus 0$ $= 0$
$(1, 2)$	$g_{1,12} = r_{0,12} \oplus r_{1,12}$ $= 0 \oplus 1$ $= 1$	$g_{2,12} = r_{0,12} \oplus r_{2,12}$ $= 0 \oplus 0$ $= 0$
$(2, 1)$	$g_{1,21} = r_{0,21} \oplus r_{1,21}$ $= 0 \oplus 1$ $= 1$	$g_{2,21} = r_{0,21} \oplus r_{2,21}$ $= 0 \oplus 1$ $= 1$
$(2, 2)$	$g_{1,22} = r_{0,22} \oplus r_{1,22}$ $= 0 \oplus 0$ $= 0$	$g_{2,22} = r_{0,22} \oplus r_{2,22}$ $= 0 \oplus 1$ $= 1$

Table 7: Reconstruction of secret bits g_1 and g_2 from first measurement outcomes using $g_{1,ij} = r_{0,ij} \oplus r_{1,ij}$ and $g_{2,ij} = r_{0,ij} \oplus r_{2,ij}$ for each pixel pair.

8.1.2 Reconstructed Matrices

Using all computed pixel values from First Outcome:

$$G_1 = \begin{bmatrix} g_{1,11} & g_{1,12} \\ g_{1,21} & g_{1,22} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$G_2 = \begin{bmatrix} g_{2,11} & g_{2,12} \\ g_{2,21} & g_{2,22} \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}$$

8.1.3 Verification of First Outcome

Comparing with the original matrices:

$$G_1^{\text{original}} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad G_2^{\text{original}} = \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}$$

We observe:

$$G_1^{\text{reconstructed}} = G_1^{\text{original}}, \quad G_2^{\text{reconstructed}} = G_2^{\text{original}}$$

Thus, reconstruction of first outcome is exact.

8.1.4 Reconstructed Matrices

Using all computed pixel values from First Outcome:

$$G_1 = \begin{bmatrix} g_{1,11} & g_{1,12} \\ g_{1,21} & g_{1,22} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$G_2 = \begin{bmatrix} g_{2,11} & g_{2,12} \\ g_{2,21} & g_{2,22} \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}$$

8.1.5 Verification of First Outcome

Comparing with the original binary images:

$$G_1^{\text{original}} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad G_2^{\text{original}} = \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}$$

We observe:

$$G_1^{\text{reconstructed}} = G_1^{\text{original}}, \quad G_2^{\text{reconstructed}} = G_2^{\text{original}}$$

Thus, reconstruction of second outcome is exact.

8.2 Given Second Outcome

$$U = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, \quad S_1 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad S_2 = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$$

These correspond to:

$$U = [r_{0,ij}], \quad S_1 = [r_{1,ij}], \quad S_2 = [r_{2,ij}]$$

8.2.1 Pixel-by-Pixel Reconstruction

Pixel Pair (i, j)	$g_{1,ij} = r_{0,ij} \oplus r_{1,ij}$	$g_{2,ij} = r_{0,ij} \oplus r_{2,ij}$
$(1, 1)$	$g_{1,11} = r_{0,11} \oplus r_{1,11}$ $= 1 \oplus 1$ $= 0$	$g_{2,11} = r_{0,11} \oplus r_{2,11}$ $= 1 \oplus 1$ $= 0$
$(1, 2)$	$g_{1,12} = r_{0,12} \oplus r_{1,12}$ $= 1 \oplus 0$ $= 1$	$g_{2,12} = r_{0,12} \oplus r_{2,12}$ $= 1 \oplus 1$ $= 0$
$(2, 1)$	$g_{1,21} = r_{0,21} \oplus r_{1,21}$ $= 1 \oplus 0$ $= 1$	$g_{2,21} = r_{0,21} \oplus r_{2,21}$ $= 1 \oplus 0$ $= 1$
$(2, 2)$	$g_{1,22} = r_{0,22} \oplus r_{1,22}$ $= 1 \oplus 1$ $= 0$	$g_{2,22} = r_{0,22} \oplus r_{2,22}$ $= 1 \oplus 0$ $= 1$

Table 8: Reconstruction of secret bits g_1 and g_2 from second measurement outcomes using $g_{1,ij} = r_{0,ij} \oplus r_{1,ij}$ and $g_{2,ij} = r_{0,ij} \oplus r_{2,ij}$ for each pixel pair.

8.2.2 Reconstructed Matrices

Using all computed pixel values from Second Outcome:

$$G_1 = \begin{bmatrix} g_{1,11} & g_{1,12} \\ g_{1,21} & g_{1,22} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$G_2 = \begin{bmatrix} g_{2,11} & g_{2,12} \\ g_{2,21} & g_{2,22} \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}$$

8.2.3 Verification of Second Outcome

Comparing with the original binary images:

$$G_1^{\text{original}} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad G_2^{\text{original}} = \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}$$

We observe:

$$G_1^{\text{reconstructed}} = G_1^{\text{original}}, \quad G_2^{\text{reconstructed}} = G_2^{\text{original}}$$

Thus, reconstruction of second outcome is exact.

8.2.4 Reconstructed Matrices

Using all computed pixel values from Second Outcome:

$$G_1 = \begin{bmatrix} g_{1,11} & g_{1,12} \\ g_{1,21} & g_{1,22} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$
$$G_2 = \begin{bmatrix} g_{2,11} & g_{2,12} \\ g_{2,21} & g_{2,22} \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}$$

8.2.5 Verification of Second Outcome

Comparing with the original matrices:

$$G_1^{\text{original}} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad G_2^{\text{original}} = \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}$$

We observe:

$$G_1^{\text{reconstructed}} = G_1^{\text{original}}, \quad G_2^{\text{reconstructed}} = G_2^{\text{original}}$$

Thus, reconstruction of second outcome is exact.

8.3 Reconstruction proof

We used:

$$g_{1,ij} = r_{0,ij} \oplus r_{1,ij}$$

Substitute:

$$r_{1,ij} = g_{1,ij} \oplus r_{0,ij}$$

$$g_{1,ij} = r_{0,ij} \oplus (g_{1,ij} \oplus r_{0,ij})$$

Using XOR associativity:

$$= (r_{0,ij} \oplus r_{0,ij}) \oplus g_{1,ij}$$

$$= 0 \oplus g_{1,ij} = g_{1,ij}$$

Similarly:

$$g_{2,ij} = g_{2,ij}$$

8.4 Conclusion

The reconstruction process successfully recovers both images exactly using:

$$G_1 = U \oplus S_1, \quad G_2 = U \oplus S_2$$

This confirms correctness of the protocol at the pixel level.

9 Security Proof

9.1 Given

$$r_1 = g_1 \oplus r_0, \quad r_0 \sim \text{Uniform}(0, 1)$$

9.2 Compute Distribution

If $g_1 = 0$:

$$r_1 = r_0 \Rightarrow P(r_1 = 0) = 1/2$$

If $g_1 = 1$:

$$r_1 = 1 \oplus r_0 \Rightarrow P(r_1 = 0) = 1/2$$

Thus:

$$P(r_1|g_1 = 0) = P(r_1|g_1 = 1)$$

$$\Rightarrow P(g_1|r_1) = P(g_1)$$

$$\Rightarrow I(g_1; r_1) = 0$$

10 Healthcare Mapping

- Alice (Patient) $\rightarrow U$
- Bob (Hospital A) $\rightarrow S_1$
- Charlie (Hospital B) $\rightarrow S_2$

no single party can reconstruct data.

11 Extension to (n, n) GHZ-Based Quantum Secret Sharing

In the previous sections, we constructed a 3-qubit GHZ-based protocol for encoding two secret bits. We now generalize this framework to an (n, n) quantum secret sharing scheme using an n -qubit GHZ state.

11.1 Generalized GHZ State

The n -qubit GHZ state is defined as:

$$|\text{GHZ}_n\rangle = \frac{1}{\sqrt{2}} (|0\rangle^{\otimes n} + |1\rangle^{\otimes n})$$

Explicitly:

$$|\text{GHZ}_n\rangle = \frac{1}{\sqrt{2}} (|000 \dots 0\rangle + |111 \dots 1\rangle)$$

11.2 Encoding Multiple Secret Bits

We encode $(n - 1)$ classical secret bits:

$$(g_1, g_2, \dots, g_{n-1}) \in \{0, 1\}^{n-1}$$

The encoding operation is defined as:

$$U_{\text{enc}} = I \otimes X^{g_1} \otimes X^{g_2} \otimes \dots \otimes X^{g_{n-1}}$$

Applying this operator:

$$|\psi_{\vec{g}}\rangle = U_{\text{enc}}|\text{GHZ}_n\rangle$$

11.3 Expanded State Form

Using tensor product properties:

$$|\psi_{\vec{g}}\rangle = \frac{1}{\sqrt{2}} (|0, g_1, g_2, \dots, g_{n-1}\rangle + |1, 1 \oplus g_1, 1 \oplus g_2, \dots, 1 \oplus g_{n-1}\rangle)$$

This is the direct generalization of the 3-qubit GHZ encoding.

11.4 Measurement Outcomes

Upon measurement in the computational basis:

$$(r_0, r_1, \dots, r_{n-1}) = \begin{cases} (0, g_1, g_2, \dots, g_{n-1}) \\ (1, 1 \oplus g_1, 1 \oplus g_2, \dots, 1 \oplus g_{n-1}) \end{cases}$$

Thus, only two outcomes have non-zero probability.

11.5 Reconstruction Rule

Each secret bit can be reconstructed using:

$$g_k = r_0 \oplus r_k \quad \forall k \in \{1, 2, \dots, n-1\}$$

11.6 Share Distribution

Each participant holds one qubit:

$$\{r_0, r_1, r_2, \dots, r_{n-1}\}$$

Reconstruction requires all participants, as:

- r_0 is required to decode all g_k
- Missing any r_k prevents recovery of the full secret.

Thus, the scheme satisfies the (n, n) threshold property.

11.7 Quantum Circuit Construction

The n -qubit GHZ state is constructed as follows:

- Apply Hadamard gate to qubit 0
- Apply CNOT gates from qubit 0 to all other qubits

Encoding is performed by applying:

$$X^{g_k} \text{ on qubit } k, \quad k = 1, 2, \dots, n-1$$

11.8 Security Intuition

Each individual qubit is maximally mixed:

$$\rho_i = \frac{I}{2}$$

Thus, no single participant can extract any information about the secret.

The secret is encoded entirely in the global correlations of the entangled state, ensuring perfect secrecy.

11.9 Separability Threshold

A subtle drawback to consider in this expansion is the Separability Threshold, which is defined as the noise level at which the state becomes fully separable.

The effective coherence of multipartite states, such as GHZ states, decays exponentially with increasing qubit number N under independent depolarising noise, causing entanglement robustness to deteriorate on an approximate inverse scaling basis. As N increases, the tolerable noise threshold before separability decreases in practical usefulness, meaning the protocol's entanglement-based security can degrade rapidly for large-scale systems.

12 Analytical Reference: GHZ-3 State Fidelity Under Noise

This section presents the analytical expressions for the fidelity of a three-qubit GHZ state under different single-channel noise models. These expressions serve as a theoretical benchmark for evaluating the robustness of GHZ-based quantum secret sharing.

12.1 General Fidelity Expression

For an N -qubit GHZ state subjected to a symmetric noise channel, the fidelity is given by:

$$F_{\text{GHZ}}^{(N)} = \frac{1}{2} \left[\left(\frac{1+s}{2} \right)^N + \left(\frac{1-s}{2} \right)^N + s^N \right]$$

where:

$$s = 1 - \lambda$$

For the specific case of a three-qubit system ($N = 3$), this expression is used to derive closed-form results under different noise models.

12.2 Noise Models and Analytical Expressions

12.2.1 Depolarizing Noise

$$F = \frac{1 + 3s^2 + 4s^3}{8}$$

where:

$$s = 1 - \lambda_{\text{Qiskit}}$$

Note that in standard textbook notation:

$$p = \frac{3}{4}\lambda$$

12.2.2 Bit-Flip Noise

$$F = (1 - p)^3 + p^3$$

This expression reflects the probability that either no qubits flip or all three qubits flip simultaneously, preserving the GHZ correlation.

12.2.3 Phase-Flip Noise

$$F = \frac{1 + (1 - 2p)^3}{2}$$

This arises due to phase coherence decay affecting the superposition components of the GHZ state.

12.2.4 Amplitude Damping Noise

$$F = \frac{1 + \gamma^3 + (1 - \gamma)^3 + 2(1 - \gamma)^{3/2}}{4}$$

This model captures energy dissipation processes, where qubits decay from $|1\rangle$ to $|0\rangle$.

12.3 Fidelity Threshold

A critical reference point for all models is:

$$F = 0.5$$

Below this threshold, the quantum state loses its non-classical advantage and behaves effectively as a classical mixture.

12.4 Visualization

The fidelity functions are plotted over the noise parameter range:

$$\lambda, p, \gamma \in [0, 0.5]$$

Each curve illustrates how different noise channels affect the stability of GHZ entanglement. The depolarizing and phase-flip channels typically degrade coherence more rapidly, while bit-flip noise preserves symmetry in certain regimes.

12.5 Interpretation

These analytical expressions demonstrate that:

- GHZ states are highly sensitive to phase-related noise
- Bit-flip errors can sometimes preserve global correlations
- Amplitude damping introduces asymmetric decay effects
- Fidelity degradation is nonlinear and depends strongly on the noise model

12.6 Conclusion

This analytical framework provides a rigorous foundation for evaluating the robustness of GHZ-based quantum secret sharing protocols under realistic noise conditions. It also establishes benchmarks for comparing experimental or simulated results.